

SQDeChain: Enhancing Tourist Trust and Destination Image Through a Hybrid AI-Blockchain Framework for Tourism 5.0

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ABSTRACT

Tourism 5.0 demands scalable, secure, and transparent systems to enhance service quality, destination image, and visitor loyalty. Traditional approaches face challenges in integrating Artificial Intelligence (AI) with blockchain due to scalability and computational costs. This paper introduces SQDeChain (SQcoin), a hybrid framework combining lightweight AI models with a sharded, Byzantine Fault-Tolerant (BFT) blockchain. Using DistilDeBERTa for text analysis, MobileViT for image processing, and GRU for behavioural predictions, the framework generates key performance indicators (KPIs), Service Quality Trust Index (SQTI), Destination Image Reliability Score (DIRS), and Tokenised Loyalty Propensity (TLP). These KPIs are verified using a novel ClusterPioneer BFT consensus mechanism and energy-efficient operation. The system was evaluated using real-world tourism datasets from Karachi, showing a 41% improvement in scalability, a 0.11 increase in predictive accuracy, and low consensus latency (400 ms). The results demonstrate that SQDeChain offers a sustainable, transparent solution for the future of smart tourism.

Keywords: Smart Tourism Systems; Tourist Satisfaction; Sentiment Analysis

1 Introduction

The advent of Tourism 5.0, characterised by increased reliance on advanced digital technologies, necessitates robust, scalable infrastructure to manage intricate data flows and ensure high standards of service delivery [10]. Within this evolving landscape, the integration of cutting-edge technologies like Artificial Intelligence, Blockchain, and the Internet of Things is crucial for fostering sustainable tourism development by addressing challenges such as environmental degradation and cultural disruption [6]. Convergence is particularly important in the development of smart tourism destinations that will use AI and IoT to provide personalised suggestions and real-time data collection to improve travellers' experiences and the overall management of destinations to their best. However, these integrated systems will be effective, but they will rely on the technology systems behind them that guarantee data integrity, transparency, and rapid processing, something centralised systems lack, providing them with an advantage over their competitors [13]. This is a constraint that indicates the need for decentralised systems, such as blockchain, that offer greater security, immutability, and transparency to better manage tourism data and transactions [1]. To be more precise, it is blockchain technology that enables secure exchanges through a decentralised network. Other than that, there are also smart contracts to automate transactions. Therefore, these technologies are helping create a safer environment for the tourism industry. It also increases operational efficiency and provides transparency [12], [8]. The framework can be useful for handling unstructured data from various sources. For example, deep learning and AI can be used to evaluate hotel reviews, extract information about their enjoyment, and classify worthwhile feedback [3].

Furthermore, customer reviews, primarily received online, have become an essential input for any hospitality company, with a direct effect on its image and financial success [11]. Despite the autonomous development of AI and blockchain in tourism, a major gap remains. It does not have a framework that provides sufficient leverage for lightweight AI to efficiently generate KPIs on the device using multimodal data and tie them to a scalable, non-mining blockchain for verification in a decentralised manner. This disconnect leads to either non-scalable, accurate but centralised systems or non-scalable, decentralised systems that are inefficient and unable to be analysed using sophisticated, verifiable analytics. This gap needs to be bridged to make Tourism 5.0 a reality. Conventional blockchain systems, such as Proof-of-Work (PoW), are energy-intensive, and Bitcoin is estimated to consume more than 150 TWh of energy each year, equivalent to the energy use of a country. The non-mining, sharded BFT of SQDeChain reduces it by more than 95%, which is why this is a feasible solution for sustainable digital infrastructure in tourism [26].

Tourism 5.0 systems must process large volumes of multimodal data, such as text reviews, images, and interaction logs, to generate reliable tourism KPIs. However, centralised AI systems lack transparency and are susceptible to data manipulation, while traditional blockchain systems struggle to scale verification of AI results. This study addresses how multimodal AI-based KPIs can be generated and verified in a decentralised, scalable, and energy-efficient environment, and proposes SQDeChain,

which combines lightweight AI models with shard-aware BFT blockchain consensus. This study evaluates four hypotheses: **H1**, integrating lightweight multimodal AI with sharded BFT blockchain consensus improves scalability compared to traditional blockchain tourism systems. **H2**, lightweight multimodal AI models can reliably generate tourism KPIs such as service quality, destination image reliability, and loyalty propensity. **H3**, blockchain-based verification of AI-generated KPIs increases transparency and trust compared to conventional off-chain AI systems. **H4**, a reputation-aware shard consensus mechanism, can maintain security against malicious validators while preserving low-latency performance.

Tourism 5.0 demands innovative digital solutions to address the growing complexity of data management, real-time personalisation, and trust. Current tourism infrastructures fail to meet these requirements, often resulting in energy-intensive systems, limited scalability, and compromised user privacy. To bridge this gap, this paper makes the following key contributions. We propose the SQDeChain (SQcoin) architecture, a novel integration of a hybrid lightweight deep learning stack with a sharded BFT blockchain, specifically designed for the tourism sector. Design the ClusterPioneer BFT consensus mechanism, which uses domain-based sharding and reputation-aware committee selection to achieve high throughput and low latency without energy-intensive mining. Introduce a novel mathematical formulation for verifiable KPIs, SQT, DIRS, and TLP that seamlessly integrates AI-derived insights with on-chain tokenomics. Conduct an extensive evaluation using both simulation and a real-world testbed with datasets from Karachi, demonstrating significant improvements in predictive accuracy, scalability, fairness, and security over baseline approaches.

The remainder of this paper is organised as follows. Section 2 reviews the literature and identifies the research gap. Section 3 presents the proposed SQDeChain framework. Section 4 describes the experimental setup and evaluation. Section 5 discusses the results. Section 6 provides the discussion, and Section 7 concludes the paper with future research directions.

2 Literature Review and Research Gap

Recent advances in smart tourism systems integrate Artificial Intelligence (AI), Internet of Things (IoT), and blockchain to enhance destination management, service personalisation, and trust in digital ecosystems. AI techniques such as sentiment analysis, image recognition, and predictive modelling are widely used to analyse tourist behaviour, while blockchain improves transparency, transaction integrity, and decentralised governance. However, most studies treat AI and blockchain as separate solutions. This creates a key limitation: centralised AI offers strong analytics but lacks transparency, whereas blockchain ensures integrity but struggles with scalability and multimodal data processing.

The current literature highlights the growing adoption of AI in smart tourism, yet there is limited work on unified frameworks that combine AI with blockchain. This gap underscores the need for integrated systems that address challenges such as data privacy, security, and fragmentation [1]. Multimodal approaches combining text, image, and temporal data improve explanatory power for constructs such as service quality and trust, but their use for verifiable KPIs remains limited [9], [10]. Lightweight hybrid models are therefore needed for deployment in mobile and edge environments.

Although blockchain has been widely explored for enhancing security and transparency in tourism [1], [8], scalability issues persist, especially in data-intensive contexts. Similarly, AI applications in tourism are rarely combined with blockchain for real-time decentralised validation [2]. Overall, existing systems largely rely on either AI or blockchain independently, revealing a critical research gap. This underscores the need for scalable hybrid architectures capable of processing multimodal data while delivering decentralised, verifiable tourism KPIs. Recent advancements in energy-efficient artificial intelligence have emphasised the importance of Green AI techniques to reduce computational overhead and environmental impact [46, 47].

High-capacity networks achieve state-of-the-art performance, but tourism applications require lightweight architectures to balance accuracy with deployment constraints. Prior work recommends distilled models, mobile vision backbones, and compact temporal networks as efficient baselines. Table 1 compares SQcoin consensus paradigms across energy, latency, fairness, shardability, and tourism adoption. Emerging approaches such as layered PBFT, shard-aware BFT, and sharding protocols demonstrate that combining BFT with sharding can preserve Byzantine resilience while improving scalability. However, these designs remain underexplored in tourism, particularly for integrating AI-derived KPIs. Cluster-leader voting offers a practical trade-off between decentralisation, efficiency, and auditability, yet its application in cluster-shard BFT for tourism systems remains limited.

Tourism analytics involves heterogeneous demand supply dynamics, requiring advanced data science techniques such as predictive modelling and sentiment analysis [8]. Large-scale studies highlight the superiority of real-time data analytics over traditional surveys, which often become outdated [8], [9]. This necessitates robust frameworks capable of processing and securing high-volume data while maintaining decentralisation, where blockchain provides clear advantages [12], [2]. Concurrently, machine learning particularly deep learning has become central to analysing online consumer perceptions and extracting actionable insights [5], [7], supporting reputation management and service quality enhancement [4].

Despite these advances, blockchain scalability, data quality, cost, and privacy issues continue to hinder adoption in data-intensive tourism systems. While recent studies integrate ML with blockchain for resilience (e.g., cyber-attack detection in smart grids [27]), challenges such as model overfitting and adaptability persist [8], [4]. Critically, there remains a lack of

Table 1. Comparison of candidate AI models for tourism analytics. The table summarises strengths and limitations of text, image, and temporal deep learning architectures, and justifies the selection of efficient lightweight models suitable for edge-based tourism applications.

Category	Technology	Strength	Limitation	Our Choice & Ratio-	Refs
Text Model	BERT/DeBERTa	Contextual understanding	Large, slow on edge	-	[11] [15]
	DistilDeBERTa	Fast, near-BERT accuracy	Slight accuracy loss	Yes - Efficient for on-device review analysis	[11] [15]
Image Model	ViT/ResNet	High accuracy	Heavy computation	-	[16] [18]
	MobileViT	Mobile-friendly, good features	Lower peak accuracy	Yes - Balances accuracy & mobile efficiency	[17]
Temporal Model	LSTM	Captures long sequences	Computationally heavy	-	[8] [10]
	GRU	Efficient, captures dependencies	Weaker on very long sequences	Yes - Optimal for edge loyalty prediction	[10]
Consensus	PoW	Secure	High energy, slow	No - Unsustainable for tourism	[24] [25]
	PoS/DPoS	Energy efficient, fast	Centralization risk	No - Fairness concerns	[11] [24]
	PBFT	Fast finality, low energy	Poor scalability	No - Doesn't scale	[13] [22]
	Hierarchical BFT	Better scalability	Complex, emerging	Foundation for our work	[22]
	ClusterPioneer BFT	Scalable, fair, low energy	Novel, less tested	Yes - Designed for tourism verification	This Work

scalable, non-mining blockchain frameworks combined with lightweight AI for decentralised KPI verification. This gap results in systems that are either accurate but centralised, or decentralised but inefficient. To address this, we propose SQDeChain, a verifiable and energy-efficient hybrid framework tailored for Tourism 5.0.

2.1 Integration Multimodal AI with Blockchain

Recent literature further reinforces the relevance of integrating multimodal AI with scalable blockchain infrastructures in smart tourism [28], [29], [30]. In tourism and hospitality research, multimodal analytics has gained strong momentum, with recent studies proposing methodological frameworks for multimodal tourism data and showing how text-image combinations influence the destination image, review usefulness, and user engagement [31]. At the blockchain layer, Recent surveys confirm that sharding remains one of the most promising approaches for improving throughput and lowering latency in decentralised systems, especially when compared with simpler consensus only modifications [32], [33]. In parallel, recent tourism studies have shown growing interest in blockchain-enabled transformation of tourism and hospitality, but they still mainly emphasize bibliometric trends [34], transaction transparency, or platform modernisation rather than verifiable AI-driven KPI computation [35]. These developments confirm the timeliness of SQDeChain, which brings together lightweight multimodal AI, shard-aware BFT consensus, and blockchain-based verification within a unified Tourism 5.0 framework.

2.2 Early Blockchain Applications in Tourism

Early research exploring blockchain applications in tourism focused primarily on improving transparency and trust in travel transactions [36], [37], [38]. Initial studies proposed blockchain-based booking platforms, decentralised identity systems for travellers, and distributed reputation mechanisms for tourism services [39].

These early systems demonstrated the potential of blockchain to improve data integrity and reduce reliance on centralised intermediaries [40]. However, most of these approaches focused on transaction management rather than data analytics [36].

In contrast, the proposed SQDeChain framework integrates blockchain verification with AI-driven tourism analytics [33],

[37]. By combining lightweight deep learning models with a scalable BFT blockchain architecture, the framework enables verifiable computation of tourism performance metrics such as service quality, destination image reliability, and visitor loyalty [41], [42].

2.3 Research Gap

Existing Tourism 5.0 systems face a critical dual challenge. Centralised AI systems offer accurate analytics but lack transparency, scalability, and trust. Decentralised blockchain systems offer improved transparency and security. However, they are often energy-intensive, exhibit limited scalability, and are not well-suited for the real-time multimodal analytics required in smart tourism environments. Therefore, a critical research gap exists in the design of hybrid AI–blockchain architectures that combine lightweight multimodal analytics with scalable consensus mechanisms. Such systems must support on device inference, verifiable KPI computation, and energy-efficient decentralised validation suitable for Tourism 5.0 applications.

2.4 Distinction from Prior AI–Blockchain Integrations

Previous attempts to combine artificial intelligence with blockchain systems have generally treated AI models as external analytics services whose outputs are stored on-chain [33], [35]. While these approaches improve data transparency, they do not provide mechanisms to verify the integrity of AI-generated results within the blockchain consensus process. The SQDeChain architecture differs in two key ways: AI-derived tourism KPIs are cryptographically hashed and validated through the ClusterPioneer BFT consensus mechanism, enabling verifiable analytics. The hybrid architecture supports edge-based inference, allowing lightweight AI models to operate on mobile or edge devices before results are verified on-chain. This integration enables a new paradigm in which AI analytics and blockchain consensus operate as a unified trust framework for Tourism 5.0 systems. This paper makes the following novel contributions:

1. Proposes SQDeChain, a hybrid framework integrating a lightweight AI stack (DistilDeBERTa + MobileViT + GRU) with a sharded BFT blockchain, specifically optimised for tourism.
2. Introduces ClusterPioneer BFT, a novel consensus mechanism using domain-based sharding and reputation-aware committee selection to achieve high throughput, low latency, and energy efficiency without mining.
3. Formulates verifiable tourism KPIs, SQTl, DIRS, and TLP that seamlessly blend AI-driven insights with on-chain tokenomics.
4. Provides extensive evaluation using real-world datasets from Karachi, demonstrating significant improvements in scalability (+41%), accuracy (+0.11 R), fairness, and energy efficiency over existing solutions.
5. Aligns with sustainability goals by reducing energy consumption by 67% compared to PoW systems, supporting SDGs 9, 11, and 12.

3 Materials and Methods

Building on the identified research gap, this section presents the design and implementation of the proposed SQDeChain framework, including its AI models, blockchain architecture, and KPI computation. SQcoin employs a hybrid deep learning stack suitable for both offline (edge) and online (cloud) modes, ensuring adaptability across diverse tourism devices. A distilled BERT-based model is used for sentiment analysis, offering low latency and mobile compatibility. A lightweight mobile vision backbone efficiently processes destination images, while a GRU model captures temporal behaviour to predict visitor loyalty. These compact models provide strong performance with minimal computational cost, making them well-suited for real-time, resource-constrained tourism applications. In this section, we describe the computation of the key performance indicators (KPIs), namely Service Quality Trust Index (SQTl), Destination Image Reliability Score (DIRS), and Tokenised Loyalty Propensity (TLP).

3.1 Proposed System Design for SQcoin

Participants (visitors, hoteliers, agencies) register as nodes and submit multimodal data (reviews, images, behavioural logs). The Deep Learning Analytics Layer processes this data using lightweight models to compute three KPIs: SQTl, DIRS, and TLP. These KPIs are hashed and validated through ClusterPioneer BFT sharded consensus in the blockchain layer. After block finalisation, smart contracts distribute SQcoin tokens based on contribution weights, while users access results via dApps. The workflow is defined in Algorithm 1.

SQDeChain integrates sustainability through energy-efficient consensus (no mining), edge-first AI (DistilDeBERTa, MobileViT, GRU), resource-aware sharding, and eco-incentivised tokenomics. The smart contract layer runs on a permissioned

blockchain (Solidity or chaincode), with core modules for reward distribution, validator management, finality, and governance. Contracts execute automatically, while heavy AI tasks run off-chain, with only hashes stored on-chain, ensuring efficiency, transparency, and auditability.

Algorithm 1: SQcoin End-to-End Workflow

Input: User multimodal data
Output: Committed block, verified KPIs, user tokens

- 1 **Phase 1: KPI Pre-calculation**
- 2 User U submits $(Text_Review, Image, Behavior_Logs)$
- 3 $SQTI_{prelim} \leftarrow \text{DistilDeBERTa}(Text_Review)$
- 4 $DIRS_{prelim} \leftarrow \text{MobileViT}(Image)$
- 5 $TLP_{prelim} \leftarrow \text{GRU}(Behavior_Logs)$
- 6 $DL_{ResultHash} \leftarrow H(SQTI_{prelim} \parallel DIRS_{prelim} \parallel TLP_{prelim} \parallel H(Raw_Data))$
- 7 **Phase 2: Transaction Propagation**
- 8 Create transaction TX containing $DL_{ResultHash}$
- 9 Broadcast TX to the network
- 10 Route TX to the domain-specific shard
- 11 **Phase 3: Consensus**
- 12 Call $\text{ClusterPioneer}(TX, DL_{ResultHash})$
- 13 **if consensus reached then**
- 14 Commit block B to shard;
- 15 **Phase 4: Token Distribution**
- 16 Trigger smart contract after finality
- 17 $Tokens_U \leftarrow \text{RewardDistributor}(SQTI_{final}, DIRS_{final}, TLP_{final})$
- 18 Transfer $Tokens_U$ to U 's wallet
- 19 **return** $Success, Block_{BHash}, Tokens_U$

3.1.1 Hybrid Deep-Learning Model Integration

SQcoin adopts a pragmatic hybrid deep learning stack explicitly designed to operate in two modes: an on-device/edge (offline-capable) mode and an online/cloud (high-capacity) mode. In the offline mode, compact models perform inference on the user device or nearby edge nodes and produce verifiable KPI summaries. The design shown in Figure 1 describes the architecture of the proposed system. In the online mode, larger models can be used optionally for periodic recalibration, ensemble boosting, and model updates. The hybrid stack is organised as three cooperating submodels that map directly to the target tourism parameters.

Figure 1 shows the architecture enabling the secure processing of multimodal tourism data (text reviews, images, and behavioural logs) and the generation of verifiable KPIs, including SQTI, DIRS, and TLP. Figure 2 shows that each shard performs BFT consensus locally, while a beacon chain coordinates cross-shard validation and block finalisation, enabling scalable and low-latency verification of AI-generated KPIs.

The SQcoin architecture uses a multi-layer framework that starts with Node Registration for participants such as visitors, hoteliers, and agencies. Data is collected in the Data Collection & Input layer, processing multimodal content like reviews and images. In the Deep Learning Analytics layer, lightweight AI models calculate key KPIs. These are then validated in the Consensus & Blockchain layer, using ClusterPioneer BFT and sharded validation for secure transaction finalisation. The Tokenisation & Reward layer distributes SQcoin via smart contracts based on weighted scores, and governance is ensured by the On-chain Governance & Monitoring layer. The Application and User Interaction layer delivers rewards and evaluations back to participants. This figure 3 illustrates the integration of three sub-models: DistilDeBERTa (for text reviews), MobileViT (for destination images), and GRU (for behavioural data) to compute the KPIs: SQTI, DIRS, and TLP. The models process multimodal data to generate reliable, verifiable results for tourism applications. The hybrid architecture combines textual, visual, and behavioural signals to produce tourism KPIs. To evaluate the contribution of each modality, an ablation study was performed by selectively removing individual models from the inference pipeline.

For textual service quality, SQcoin uses a distilled transformer encoder, such as DistilDeBERTa, to process reviews, survey responses, and metadata, and outputs a sentiment polarity and a service-aspect embedding. These are combined with credibility features to compute an individual review score q_i , and the SQTI is calculated as:

$$SQTI = \frac{\sum_i w_i \cdot q_i}{\sum_i w_i},$$

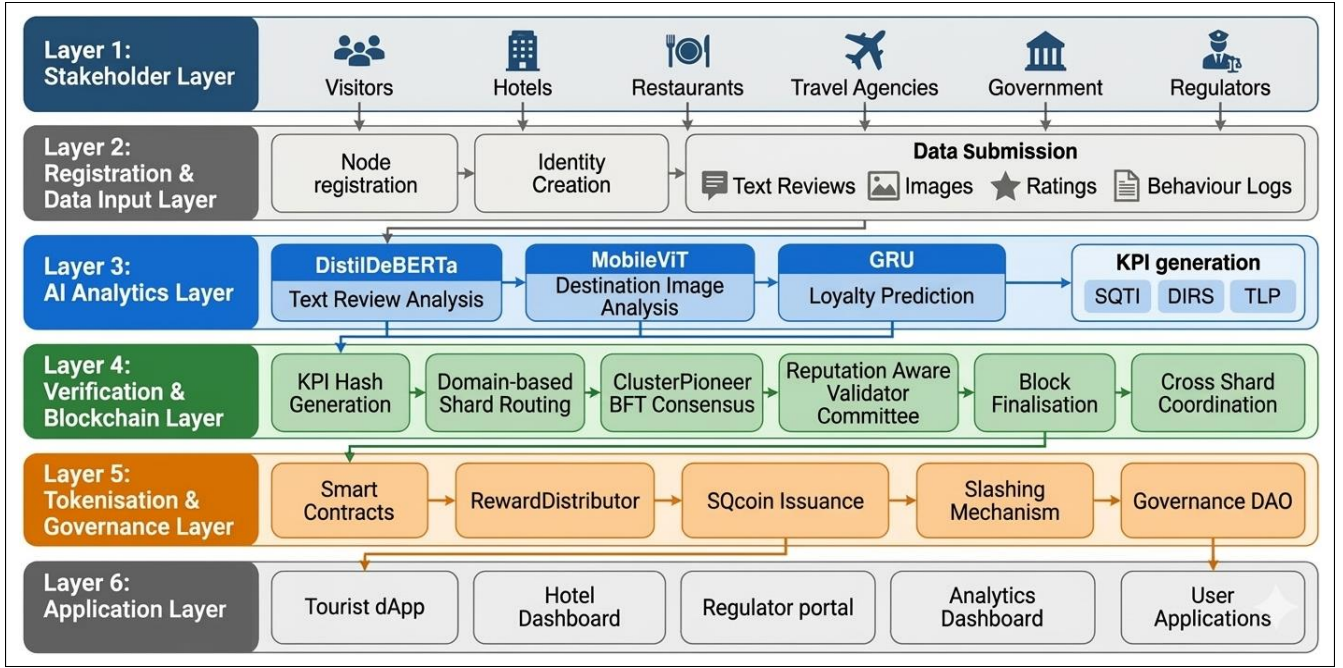


Figure 1. Overall architecture of the proposed SQDeChain (SQcoin) framework for Tourism 5.0. The system integrates six layers: stakeholder interaction; node registration and data submission; multimodal AI analytics; blockchain verification via the ClusterPioneer BFT consensus mechanism; tokenisation and governance; and application services.

where w_i reflects trust and recency. For destination image reliability, MobileViT extracts visual embeddings, and a model like CLIP aligns them with text. DIRS is computed as

$$\text{DIRS} = \frac{1}{N} \sum_{i=1}^N \cos(v_i(t_i))$$

weighted by image provenance. For loyalty, a GRU model processes user history and engagement to predict loyalty, with the final TLP given by

$$\text{TLP}_{\text{final}} = \sigma(\alpha \cdot \text{SQTl} + \beta \cdot \text{DIRS} + \gamma \cdot p_i),$$

where σ is a sigmoid. Models are versioned and identified cryptographically. For offline use, quantised models produce deterministic results, while online models run on edge/cloud oracles for recalibration and re-estimation.

3.1.2 BFT ClusterPioneer Shard Architecture

SQcoin's consensus stack is a shard-aware BFT architecture with ClusterPioneer. The blockchain is divided into shards based on tourism domains, each maintaining its own state and mempool. Validators, selected by a weighted reputation-aware lottery, are assigned to committees of size $k = 7 - 21$. A proposer aggregates transactions and the DL Result bundle, including the DL Result-Hash, into a block header. Validators perform deterministic checks, and after collecting $2f + 1$ PREPARE signatures, the leader aggregates them into a BLS signature and broadcasts a COMMIT. The finality proof is sent to the Beacon for cross-shard finality.

The Beacon coordinates epoch changes and cross-shard receipts, ensuring interactions rely on shard finality. Committee selection uses deterministic randomness (VRF), weighted by effective stake and reputation. Multi-attestation is required for DL-Result-Hash validation, with committee members re-computing or verifying proofs. If mismatches exceed a threshold, the block enters re-run or slashing. Provenance metadata is stored for audits. The FinalityContract accepts BLS aggregated signatures, using off-chain verification with zero-knowledge proofs to save on-chain verification costs. The Beacon ensures proof availability across shards.

3.1.3 SQcoin Coinchain: Verifier, Trust Builder, and Tokenomics

The SQcoin is a tokenised trust economy that validates AI-based KPIs and rewards them through smart contract distribution, governance, and enforcement. SQTl is an aggregate weighted average of review scores; DIRS is a reliability-weighted average

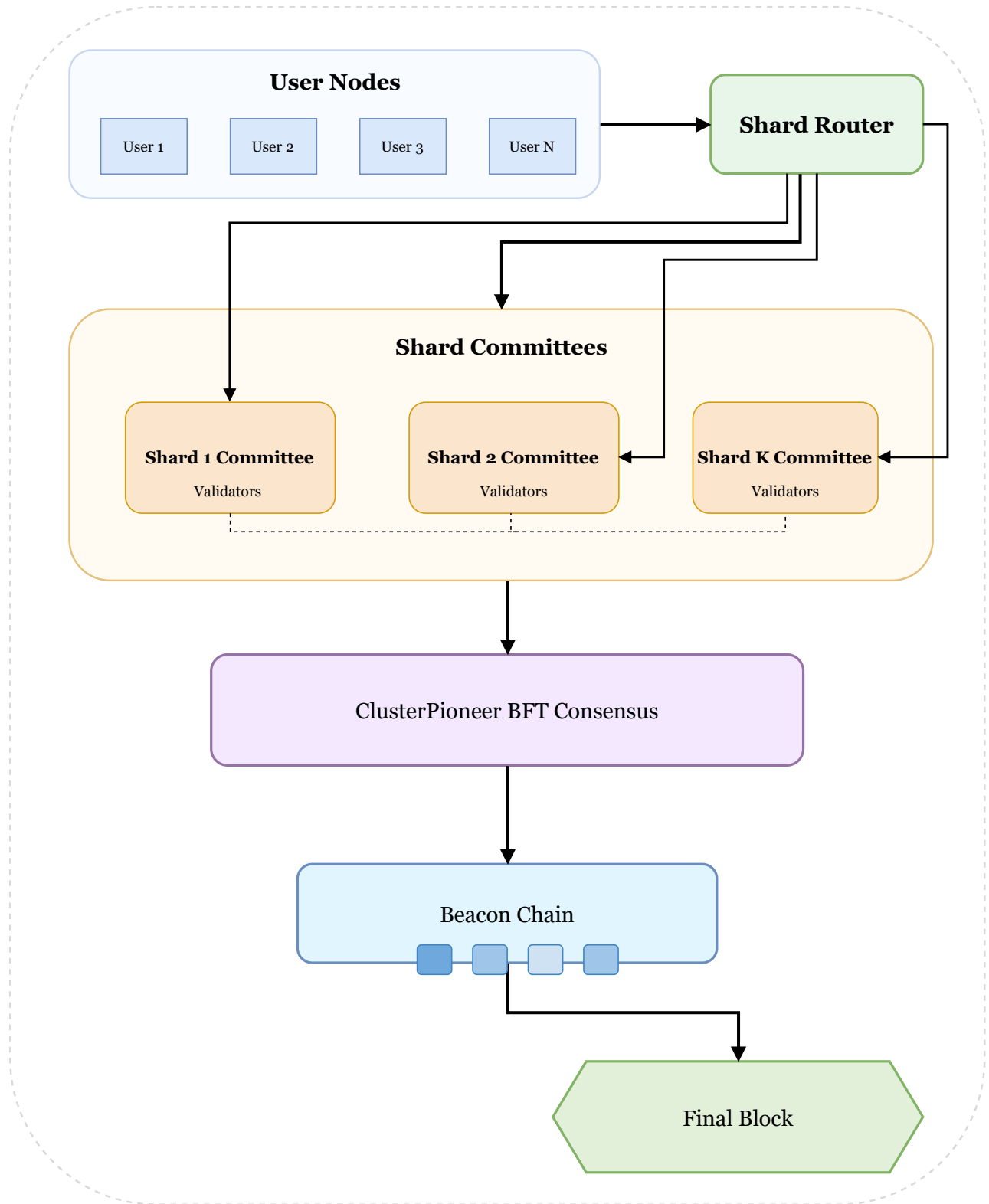


Figure 2. ClusterPioneer shard-based consensus architecture in SQDeChain. User transactions are submitted by distributed nodes and routed through a shard router to domain-specific shard committees. Each shard contains a group of validators that perform BFT consensus locally. Within each shard, validators exchange messages to verify transactions and generate block proposals.

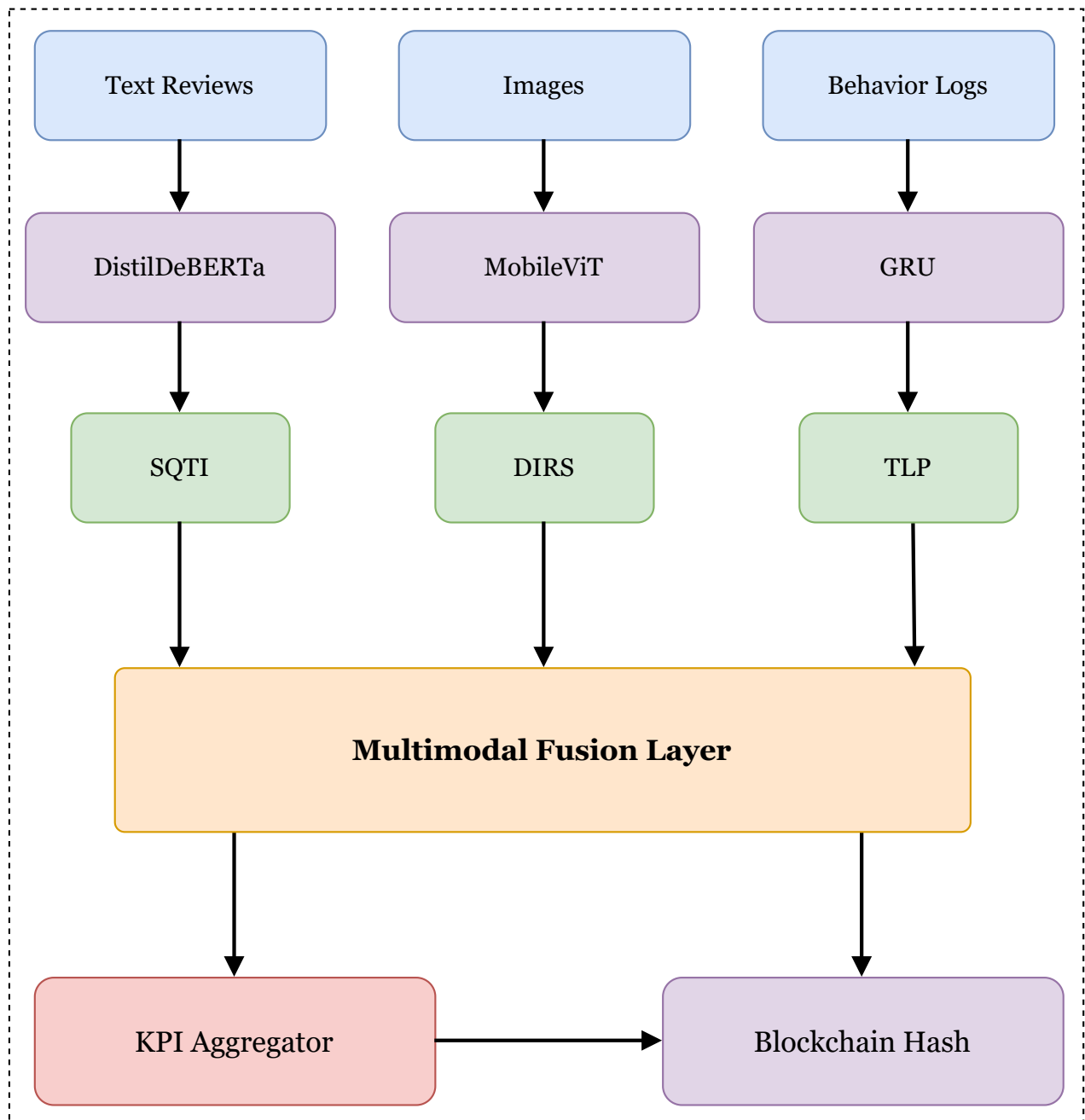


Figure 3. Shows the multimodal deep learning architecture for KPI computation in SQDeChain. It processes text reviews, destination images, and behavioural logs using DistilDeBERTa, MobileViT, and GRU models. The outputs are fused to compute three KPIs: SQT, DIRS, and TLP. These KPIs are then hashed for blockchain verification, enabling efficient and verifiable data integration.

of image-text alignment; and TLP is a combination of propensity models for token allocation. The concept of tokenomics does not involve mining, and it offers rewards to validators, visitors, and service providers. Equivocation, DL mismatches or downtime is slashed. The chain reveals measurements such as throughput, consensus latency, and KPI confidence intervals, as well as auditable off-chain datasets, which are available to verify the outputs of third-party DL models.

3.2 SQDeChain Architecture Components

SQcoin's architecture includes shards, ClusterPioneer committees, Beacon/Coordination, DL nodes/oracles, light clients, and smart contracts. Shards are geographically and domain-partitioned for efficient data and DL inference processing. Committees, typically sized $k = 11$ with $f = 3$, ensure low-latency LBFT operation. The Beacon coordinates epoch changes, finality, and aggregates BLS signatures for consensus. DL nodes, either validators with ML capabilities or dedicated oracles, ensure transparent provenance with signed, merkleized DL results. Light clients allow visitors to claim tokens via Merkle and finality proofs.

Smart contracts (ValidatorRegistry, EpochManager, FinalityContract, RewardDistributor, Token, and GovernanceDAO) ensure deterministic transaction flow. Shards handle transaction locality and DL inference, with asynchronous cross-shard receipts confirmed by Beacon finality. Committee selection combines stake and reputation, and voting uses $2f + 1$ LBFT with BLS aggregation. View changes and proposer rotation prevent faults. The BFT flow is optimised for low latency, with re-execution or proof verification by committee members for DL result validation. Slashing ensures accountability for mismatches. Tokenomics exclude mining, with rewards and inflation funding validators and visitors. Governance and slashing rules maintain long-term system soundness.

3.3 Metrics, Factors, and Measurement

SQcoin evaluates performance using operational, KPI, fairness, and security metrics. Operational metrics include throughput, latency, block size, DL processing time, validator participation, and slashing rate. KPI metrics cover SQTI, DIRS, TLP, provenance scores, and DL pass rates. Fairness is assessed via token distribution (Gini coefficient), contribution-reward balance, and provider royalties, while security ensures resistance to poisoning and correct cross-shard validation. SQTI uses bootstrap confidence limits, DIRS measures image authenticity, and TLP is validated using ROC/PR analysis. A lower Gini coefficient (0 = equal, 1 = unequal) indicates fairer rewards, which is essential for trust, participation, and sustainability.

3.4 Reputation-Aware Validator Selection

To enhance the fairness, security, and sustainability of the ClusterPioneer BFT consensus mechanism, we introduce a Reputation-Aware Validator Selection (RACS) algorithm. Unlike conventional stake-based or random selection methods, RACS incorporates multi-dimensional reputation metrics to form more reliable and geographically diverse consensus committees, thereby improving system resilience and reducing energy waste associated with malicious or unreliable validators. A validator's reputation score $R(v)$ is computed periodically per epoch and incorporates. Validation Accuracy $A(v)$: Percentage of correctly verified KPI bundles. Uptime Ratio $U(v)$: Availability during consensus rounds. Slashing History $S(v)$: Penalties for malicious acts. Geographical Distribution: Ensures decentralisation and reduces regional dominance. The composite reputation score is calculated as:

$$R(v) = \frac{\sum_{i=1}^t \gamma^{t-i} \cdot A_i(v)}{\sum_{i=1}^t \gamma^{t-i}} + \beta \cdot \log(1 + U(v)) - \lambda \cdot S(v)$$

where γ is a time-decay factor, and β, λ are tunable weights.

The RACS algorithm 2 selects k validators for each shard committee using a weighted score that combines normalised stake and reputation:

$$\text{Score}(v) = \alpha \cdot \text{StakeNorm}(v) + (1 - \alpha) \cdot R_{\text{norm}}(v)$$

Validators with recent slashing events are penalised, while those with high uptime and accuracy are prioritised. Geographical diversity is enforced by capping the number of validators per region per committee. The RACS algorithm 2 is designed to select reliable validators for shard committees while mitigating the influence of malicious or unreliable nodes. Each validator is assigned a composite score based on normalised stake and historical reputation. Reputation is derived from validation accuracy, uptime ratio, and slashing history. Validators with higher composite scores are prioritised during committee formation, while geographic diversity constraints are enforced to prevent regional concentration of consensus power.

Algorithm 2: Reputation-Aware Committee Selection

Input: Candidate validators V , stake $S[v]$, reputation $R[v]$, committee size k **Output:** Selected committee C

```
1 Initialise empty priority queue  $Q$ ;  
2 foreach validator  $v \in V$  do  
3    $\text{score}[v] \leftarrow \alpha \cdot \text{normalize}(S[v]) + (1 - \alpha) \cdot R[v]$ ;  
4   if  $v.\text{slashed\_in\_last\_epoch}$  then  
5      $\text{score}[v] \leftarrow \text{score}[v] \cdot \text{penalty\_factor}$ ;  
6    $\text{uptime\_bonus} \leftarrow \log(1 + v.\text{uptime\_ratio})$ ;  
7    $\text{score}[v] \leftarrow \text{score}[v] \cdot (1 + \beta \cdot \text{uptime\_bonus})$ ;  
8   Insert  $(\text{score}[v], v)$  into  $Q$ ;  
9  $C \leftarrow []$ ;  
10  $\text{regions\_covered} \leftarrow \{\}$ ;  
11 while  $|C| < k$  and  $Q$  not empty do  
12    $v \leftarrow \text{pop\_max}(Q)$ ;  
13   if  $\text{regions\_covered}[v.\text{region}] < \text{max\_per\_region}$  then  
14     Add  $v$  to  $C$ ;  
15      $\text{regions\_covered}[v.\text{region}] ++$ ;  
16 if  $|C| < k$  then  
17   Fill remaining slots with top validators from  $Q$ ;  
18 return  $C$ ;
```

3.5 Consensus Complexity Analysis

The ClusterPioneer BFT protocol has the communication complexity characteristic of standard BFT protocols based on shards (Table 2). The consensus protocol needs message exchanges of $O(k^2)$ in each shard committee of size k . The size of each committee, however, is limited by sharding, which enables the system to expand horizontally as the number of nodes increases. Since consensus operations are independent across shards, the total network throughput increases with the number of shards, and everything is built with Byzantine fault tolerance guarantees. In contrast to conventional BFT protocols, which use the entire set of validators, the ClusterPioneer architecture executes consensus on small shard committees, enabling higher throughput and scalability for tourism data loads.

Table 2. Communication complexity and fault tolerance comparison of Byzantine Fault Tolerant consensus protocols.

Protocol	Communication Complexity	Scalability	Fault Tolerance
PBFT	$O(n^2)$	Limited	Byzantine
HotStuff	$O(n)$ (linear view change)	Moderate	Byzantine
Tendermint	$O(n^2)$	Moderate	Byzantine
ClusterPioneer BFT	$O(k^2)$ (per shard)	High	Byzantine

3.6 Recommended Defaults and Tuning Values

Recommended configuration: 5-second shard block interval, 100–300 event batches, committee size $k = 11$ ($f = 3$), epoch length of 1,000 blocks, and 2–4 initial shards. Minimum stake ensures 100–300 validators without a Sybil attack, with 1–2% annual inflation. DL verification threshold m should be at least $\lceil k/2 \rceil$, higher for high-value KPI batches. Cross-shard receipt acceptance requires Beacon finality with an `agg_sig` verification window of one epoch. Figure 4 visualise the token distribution, slashing, and governance mechanisms.

3.7 Security Model and Assumptions

The security of SQDeChain is analysed under a Byzantine adversarial model with a partially synchronous network. The system includes users, validator nodes, and smart contracts. Users are untrusted, validators are semi-trusted with $f \leq (n - 1)/3$, and smart contracts are deterministic and trusted after deployment. The adversary may launch Sybil, data poisoning, Byzantine, and collusion attacks, but does not control the majority of the network. Cryptographic primitives (e.g., hashing and signatures) are assumed to be secure. ClusterPioneer BFT ensures security via multi-attestation and slashing, while reputation-based validator selection limits adversarial influence.

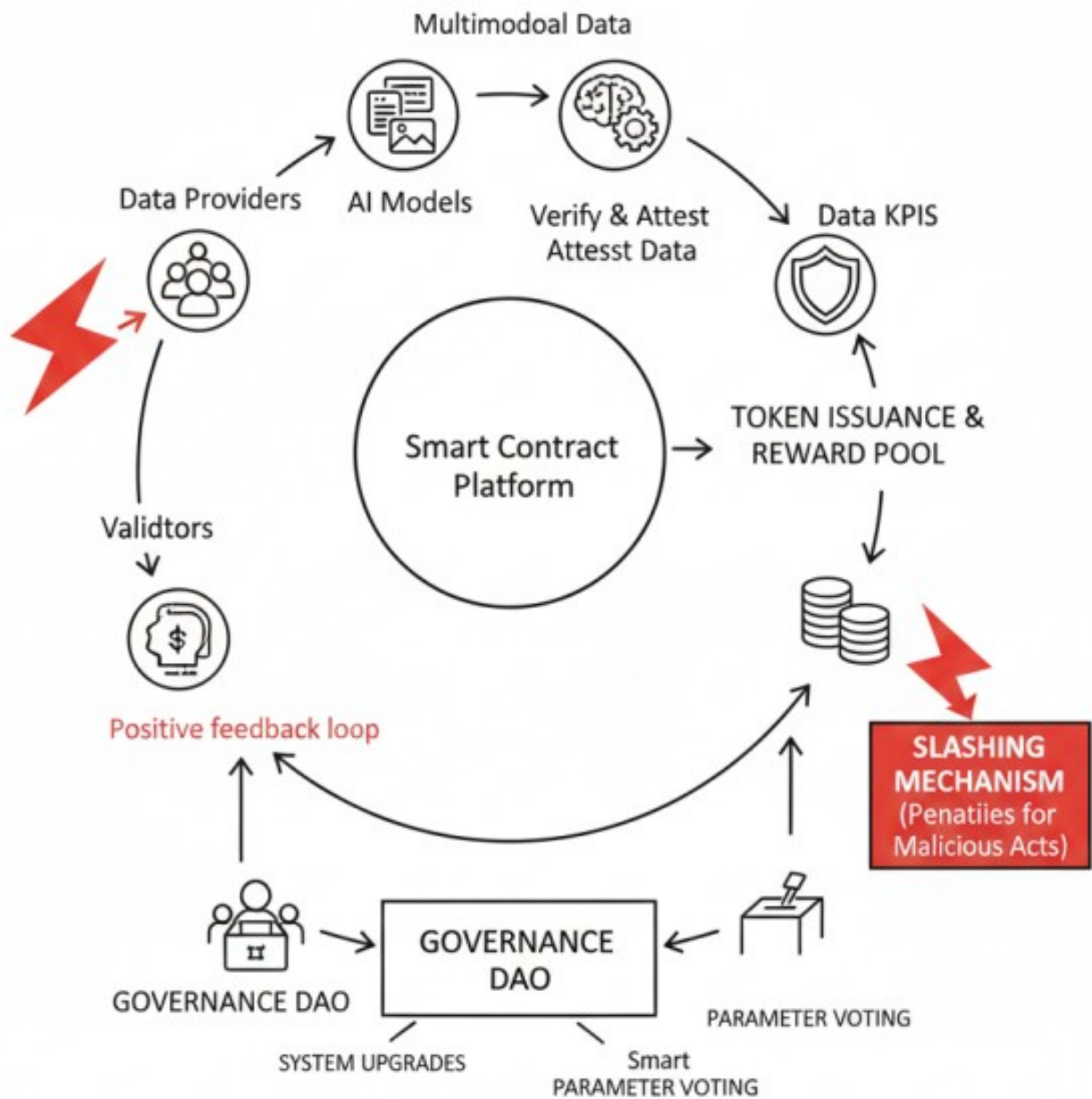


Figure 4. Tokenomics and governance model of the SQcoin ecosystem. AI-generated tourism KPIs are verified via blockchain consensus and submitted to smart contracts that distribute token rewards from a reward pool.

3.8 Validator Onboarding and Removal Process

The SQDeChain framework defines a validator lifecycle comprising onboarding, active participation, and removal to ensure security and reliability.

Validator Onboarding: Validators register via the ValidatorRegistry smart contract by submitting credentials, stake (if applicable), and configuration details. They receive an initial reputation score and are added to a candidate pool. Selection into shard committees is performed per epoch using the Reputation-Aware Committee Selection (RACS) algorithm, which is based on stake and reputation.

Active Participation: Validators validate transactions and verify KPI hashes. Performance is tracked through accuracy, uptime, and protocol compliance, with reputation scores updated dynamically.

Validator Removal and Penalties: (i) Malicious actions (e.g., conflicting votes, invalid signatures) trigger slashing and exclusion; (ii) low uptime reduces reputation and selection probability; (iii) repeated violations lead to blacklisting. Slashing is enforced via smart contracts, ensuring transparency and maintaining consensus integrity.

3.9 Mathematical Derivations of SQDeChain

3.9.1 Service Quality Trust Index

SQTI is defined as the weighted average of review-level quality scores:

$$SQTI = \frac{\sum_{i=1}^n w_i q_i}{\sum_{i=1}^n w_i} \quad (1)$$

where:

- n is the total number of review instances,
- $q_i \in [0, 1]$ is the normalized quality score of review i ,
- $w_i > 0$ is the adaptive weight assigned to review i , reflecting factors such as reviewer credibility, recency, and platform trust.

The weights w_i are constrained such that:

$$w_i > 0, \quad \sum_{i=1}^n w_i = 1$$

ensuring that SQTI represents a normalised weighted average of review quality scores. This formulation incorporates review credibility, recency, and trustworthiness, allowing more reliable reviews to have greater influence on the overall service quality estimate.

To reduce sensitivity to outliers, a smoothed variant is used:

$$SQTI^* = \alpha \cdot SQTI + (1 - \alpha) \cdot \text{Median}(q_i), \quad 0 < \alpha < 1 \quad (2)$$

where α is a smoothing parameter that balances the weighted mean with the median review quality. This formulation was adopted to preserve aggregate sentiment information while reducing the effect of anomalous or manipulated reviews.

3.9.2 Destination Image Reliability Score (DIRS)

DIRS evaluates the trustworthiness of user-generated visual and textual content. It is computed as a weighted combination of image similarity, text sentiment polarity, and user credibility.

$$DIRS = \frac{1}{m} \sum_{j=1}^m (\beta_1 \cdot \text{Sim}(I_j, I^*) + \beta_2 \cdot \text{Pol}(T_j) + \beta_3 \cdot \text{Ver}(U_j))$$

where $\text{Sim}(I_j, I^*)$ represents the cosine similarity between the uploaded image and the reference destination embedding, $\text{Pol}(T_j)$ represents the sentiment polarity of the associated textual review, and $\text{Ver}(U_j)$ denotes the credibility score of the submitting user.

The weighting parameters satisfy the constraint

$$\beta_1 + \beta_2 + \beta_3 = 1$$

ensuring balanced contribution of visual, textual, and reputation-based signals.

3.9.3 Validator Reputation Model

The reputation score of a validator v is defined as

$$R(v) = \frac{\sum_{i=1}^t \gamma^{t-i} A_i(v)}{\sum_{i=1}^t \gamma^{t-i}} + \beta \cdot \log(1 + U(v)) - \lambda \cdot S(v)$$

where $A_i(v)$ denotes validation accuracy during epoch i , $U(v)$ represents validator uptime, and $S(v)$ indicates the number of slashing events.

The parameter t denotes the number of historical epochs considered, while the decay parameter γ satisfies

$$0 < \gamma < 1$$

ensuring that recent validator behaviour has greater influence than older observations.

3.9.4 Tokenised Loyalty Propensity

TLP models the likelihood that a visitor will demonstrate repeat engagement and therefore qualify for token-based incentives:

$$TLP_k = \gamma_1 Freq_k + \gamma_2 Dur_k + \gamma_3 SQT I_k^* + \gamma_4 DIRS_k \quad (5)$$

where:

- TLP_k is the loyalty propensity score of visitor k ,
- $Freq_k$ is the normalized visit frequency,
- Dur_k is the normalised visit duration or engagement time,
- $SQT I_k^*$ is the smoothed service quality score associated with visitor k ,
- $DIRS_k$ is the destination image reliability score relevant to visitor k ,
- $\gamma_1, \gamma_2, \gamma_3, \gamma_4$ are weighting parameters.

The token allocation is then bounded by:

$$Tokens_k = \min(\lambda \cdot TLP_k, T_{\max}) \quad (6)$$

where λ is a reward scaling factor and $T_{\max}$ is the maximum token cap per epoch. This prevents excessive reward concentration and supports fairness in token distribution.

3.9.5 BFT Fault Tolerance and Quorum Conditions

For a shard committee of size n , the maximum number of Byzantine validators that can be tolerated is:

$$f = \left\lfloor \frac{n-1}{3} \right\rfloor \quad (7)$$

where f denotes the number of faulty or malicious validators.

To guarantee safety and liveness under Byzantine assumptions, a valid commit requires a quorum of at least:

$$Q = 2f + 1 \quad (8)$$

validator approvals.

Thus, a block B_r proposed at round r is committed only if at least $2f + 1$ validators produce valid signatures for that block. This follows the standard safety condition of Byzantine Fault Tolerant consensus, ensuring that conflicting blocks cannot both be finalized when $n \geq 3f + 1$.

3.9.6 Aggregated Signature Condition

Let $\{\sigma_1, \sigma_2, \dots, \sigma_Q\}$ denote the set of validator signatures collected for a proposed block. The aggregated Boneh–Lynn–Shacham (BLS) signature is defined as:

$$\sigma_{agg} = \bigoplus_{i=1}^Q \sigma_i \quad (9)$$

where $Q \geq 2f + 1$ and $\bigoplus$ denotes the aggregation operator over the individual validator signatures.

The aggregated signature serves as the compact proof that the required Byzantine quorum has approved the block.

3.9.7 Cross-Shard Validity

A cross-shard receipt R_s is considered valid if the beacon layer successfully verifies the associated proof bundle:

$$\text{Valid}(R_s) \iff \text{BeaconVerify}(\text{Proofs}_s) = 1 \quad (10)$$

where Proofs_s denotes the cryptographic proof set submitted for shard s . This condition ensures that cross-shard state transitions are accepted only after finality has been established by the coordination layer.

3.9.8 Integrated Objective of SQcoin Chain

The integrated utility of the SQcoin chain is:

$$CU = \delta_1 \cdot \overline{SQTI^*} + \delta_2 \cdot \overline{DIRS} + \delta_3 \cdot \overline{TLP} - \delta_4 \cdot \text{Lat} + \delta_5 \cdot \text{Scal} \quad \text{eq. (10)}$$

The mathematical formulations were designed to align the analytical and blockchain components of SQDeChain. Equations (1)–(3) define the three AI-derived tourism KPIs using weighted multimodal aggregation, while Equations (7)–(10) formalise the consensus requirements for secure validation and cross-shard finality. Together, these expressions provide a unified mathematical view of how tourism analytics are generated, verified, and committed within the proposed framework. The proposed SQDeChain framework combines lightweight multimodal AI with a scalable shard-based BFT blockchain for verifiable tourism analytics. While its design indicates advantages in scalability, transparency, and efficiency, empirical validation is required. The following section presents a comprehensive evaluation of the system, examining AI predictive accuracy, blockchain scalability, token distribution fairness, and robustness under adversarial conditions, using both simulations and real-world testbed experiments.

4 Experiments

Building upon the proposed SQDeChain architecture described in the previous section, this section presents the experimental evaluation of the framework. Testing of the SQcoin framework is conducted in the testbed simulation and in the testbed environment. The experiments are expected to establish the system’s scalability, predictive ability, fair token distribution, and resistance to adversarial attacks. It evaluates outcomes against a set of KPIs. The experiments have been done in two stages:

Simulation Setup: A controlled environment using synthetic and publicly available datasets was set up to test the scalability, accuracy, and consensus performance of SQcoin. The simulation tests included varying the number of shards and validators to measure throughput, latency, and consensus latency.

Real-Time Testbed: A real-world testbed was deployed to evaluate the performance of the SQcoin framework under realistic conditions. The testbed included multiple nodes distributed geographically, representing visitors, service providers, and validators.

Every deep learning model was trained on an NVIDIA A100 graphics card with 40 GB of memory. The experiments were carried out in Python and PyTorch, and auxiliary libraries included HuggingFace Transformers for language models and OpenCV for image processing. The blockchain simulation system was run on a distributed CPU-based node architecture to simulate the performance of the validator node in realistic network conditions. Even though the consensus latency of the ClusterPioneer BFT mechanism is fairly constant with respect to shard configurations, overall block finality time is slightly increased in relation to the size of shards.

The reason for this behaviour is that cross-shard coordination and beacon chain verification lead to additional communication overhead. Nevertheless, the trend towards increased use is still within acceptable limits for real-time tourism analytics applications.

4.1 Experimental Design Overview

The experimental evaluation was conducted in two stages: a controlled simulation and a geographically distributed real-world testbed. The simulation enabled parameter variation (e.g., shards, committee size), while the testbed assessed performance under latency and heterogeneous hardware. This approach captures both theoretical scalability and practical constraints.

All experiments were repeated five times with different initialisations and splits, and the results are reported as mean $\pm$ SD. A 95% CI was computed via bootstrap. Paired t-tests with Bonferroni correction ($\alpha' = \alpha/m$) were used, with significance at $p < 0.05$.

4.2 Training Configuration and Hyperparameters

To ensure reproducibility, all models were trained using explicitly defined hyperparameters and dataset configurations. The experimental setup is summarised in Table 3. All datasets were divided into training (70%), validation (15%), and testing (15%) subsets using stratified sampling for the textual dataset to preserve class distribution.

Table 3. Training Hyperparameters for Different Modalities

Model	Parameter	Value
DistilDeBERTa (Text Model)	Optimizer	AdamW
	Learning rate	2×10^{-5}
	Batch size	32
	Number of epochs	5
	Maximum sequence length	256 tokens
MobileViT (Image Model)	Optimizer	Adam
	Learning rate	1×10^{-4}
	Batch size	64
	Number of epochs	20
	Input resolution	224×224
GRU (Behavioural Model)	Hidden units	128
	Sequence length	30
	Dropout rate	0.3
	Optimizer	RMSProp
	Learning rate	1×10^{-3}

Hardware and framework: NVIDIA A100 (40 GB), PyTorch, HuggingFace Transformers, OpenCV. All experiments were conducted under consistent settings to ensure comparability across models.

4.3 Simulation vs Real-World Testbed Distinction

The SQDeChain framework is evaluated using two complementary environments: a simulation setup and a real-world testbed, each serving distinct purposes. The simulation environment provides a controlled setting where parameters such as shard count, validator nodes, and transaction load are varied to measure scalability, throughput, and consensus performance under ideal conditions. In contrast, the real-world testbed reflects practical deployment with distributed nodes, network latency, heterogeneous hardware, and bandwidth constraints, enabling evaluation under realistic operational conditions. While simulation results indicate theoretical scalability and upper-bound performance, testbed results confirm practical feasibility and robustness. Differences in throughput and latency are mainly due to network variability and infrastructure heterogeneity in real-world deployment.

4.4 Experimental Plan for Karachi Demonstration

The Karachi demonstration combines simulation and live testing. Synthetic workloads and historical UGC datasets simulate shard throughput, DL processing, and consensus latencies. Baseline and hybrid DL models are evaluated on Karachi data for R^2 improvements and calibration performance. Attacker models simulate poisoning and equivocation risk under different malicious fractions and committee sizes. The live testbed uses geographically distributed validators and edge nodes for DL inference on real UGC, measuring latency, token fairness, and robustness under network jitter.

4.5 Datasets

The SQDeChain framework was evaluated across three data categories: textual reviews, visual destination images, and behavioural interaction logs. The textual, visual, and behavioural datasets were aligned with the three analytical components of

SQDeChain, namely DistilDeBERTa for review sentiment analysis, MobileViT for destination image evaluation, and GRU for loyalty propensity prediction.

4.5.1 Textual Data

Hotel and tourism review datasets from platforms such as TripAdvisor, Yelp, and Kaggle were used to train the DistilDeBERTa model for sentiment analysis. A corpus of 125,000 hotel reviews from TripAdvisor and Yelp, focusing on the Karachi region. The sentiment scores (on a scale of 1-5) had a mean of 3.8 with a standard deviation of 1.2.

4.5.2 Visual Data

50,000 geo-tagged images from Instagram and Flickr labelled for destination attributes. Used for evaluating the DIRS with the MobileViT model.

4.5.3 Behavioral Data

Synthetic booking logs for 10,000 simulated visitors, containing sequences of visit frequency, duration, and spending, were used to generate a loyalty propensity score (mean=0.65, std=0.22).

4.6 Model Training and Hyperparameter Configuration

The deep learning models used in SQDeChain were trained using domain-specific tourism datasets and fine-tuned to extract multimodal insights from textual reviews, images, and behavioural logs.

DistilDeBERTa (Text Sentiment Analysis): The model was fine-tuned using 125,000 tourism reviews collected from TripAdvisor and Yelp datasets. Text preprocessing included tokenisation, stopword removal, and normalisation. The model was trained using the AdamW optimiser with a learning rate of 2×10^{-5} . A batch size of 32 was used during training, and the model was trained for 5 epochs. The maximum sequence length was set to 256 tokens to efficiently process textual inputs.

MobileViT (Image Feature Extraction): MobileViT was trained on 50,000 geo-tagged tourism images using transfer learning from ImageNet pretrained weights. The model was trained using the Adam optimiser with a learning rate of 1×10^{-4} . A batch size of 64 was used during training, and the model was trained for 20 epochs. All images were resized to 224×224 before being fed into the network.

GRU (Behavioural Prediction Model): The GRU network processes sequential visitor data, including visit frequency, duration, and engagement patterns. The model was trained using a configuration consisting of 128 hidden units and a sequence length of 30 to capture sequential patterns in the input data. A dropout rate of 0.3 was applied to reduce overfitting and improve generalisation. The training process used the RMSProp optimiser with a learning rate of 1×10^{-3} to efficiently update the model parameters.

The tourism review dataset had class imbalance, with positive reviews much more frequent than negative ones. To reduce bias in sentiment classification, weighted cross-entropy loss, stratified batch sampling, and data augmentation for minority classes were applied. These methods helped balance the model's performance across sentiment classes and improved the reliability of SQTI.

The behavioural dataset consisted of synthetic booking and visit logs for 10,000 simulated tourists. Each sequence contained variables such as visit frequency, session duration, spending patterns, and engagement history. These logs were generated to reflect realistic tourism usage behaviour and were used to train the GRU-based loyalty prediction model.

For model development, the datasets were partitioned into training, validation, and testing subsets. Unless otherwise stated, a split ratio of 70% / 15% / 15% was used for training, validation, and testing, respectively. Stratified sampling was applied to the textual sentiment dataset to preserve label distribution across all subsets.

4.7 Adversarial Testing Scenarios

To evaluate system robustness, three adversarial attack models were simulated:

Sybil Attack: Malicious entities created multiple fake identities, attempting to influence validator selection.

Data Poisoning Attack: Adversarial users submitted manipulated reviews or images designed to bias the AI inference models.

Byzantine Validator Attack: Malicious validators attempted to disrupt consensus by submitting conflicting votes during shard consensus rounds.

Experiments were conducted with different malicious fractions, including the Byzantine threshold scenario of $f = 3$ malicious nodes within a committee size $k = 11$.

4.8 Evaluation Metrics

The SQcoin framework's performance was evaluated using key metrics: accuracy, scalability, fairness, and security. These metrics ensure the system operates efficiently and reliably, particularly in the tourism sector. Definitions and mathematical formulations for each metric are provided below.

Accuracy:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Scalability:

$$\text{Throughput (TPS)} = \frac{\text{Number of Transactions Processed}}{\text{Time Interval (seconds)}}$$

$$\text{Latency} = \text{Time taken to process a transaction (ms)}$$

Fairness:

$$G = \frac{1}{2n^2\mu} \sum_{i=1}^n \sum_{j=1}^n |x_i - x_j|$$

Security:

$$\text{Resilience (Sybil)} = 1 - \frac{\text{Successful Attacks}}{\text{Total Attempts}}$$

$$\text{Resilience (Poisoning)} = \frac{\text{Accuracy after Poisoning}}{\text{Accuracy before Poisoning}} \times 100$$

These metrics serve as the basis for the quantitative evaluation of the SQcoin framework, providing insights into its effectiveness, scalability, fairness, and security in a real-world deployment.

5 Results and Analysis

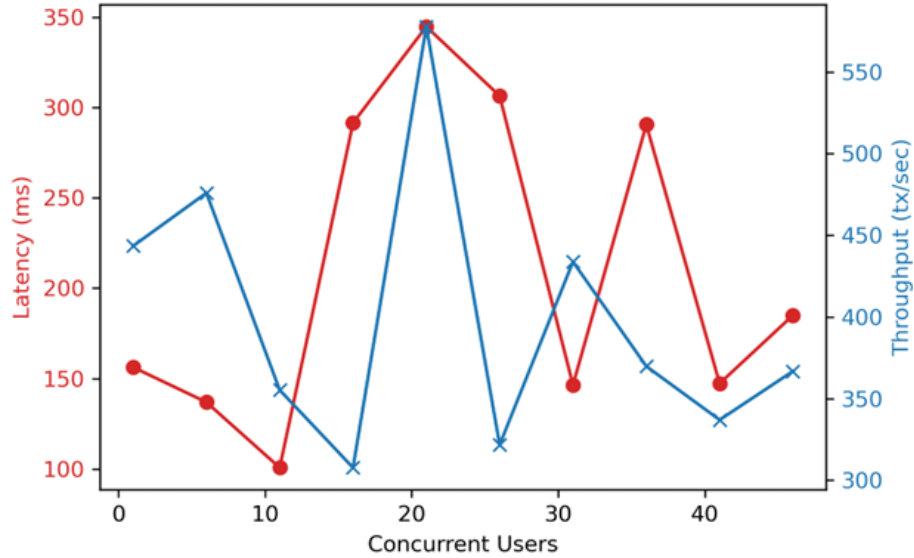
This section presents a comprehensive evaluation of the SQDeChain framework, analysing its performance across scalability, predictive accuracy, fairness, security, and energy efficiency. Results are derived from both simulation environments and a real-world testbed deployment, with comparisons against state-of-the-art baselines. All experiments were conducted using the datasets and configurations described in Section 4.

5.1 Simulation Results

To assess scalability, we simulated SQDeChain with varying numbers of shards (2–8) and validators (14–56). Throughput (transactions per second, TPS) and consensus latency were measured under a synthetic workload of 10,000 KPI submissions.

Table 4. Scalability and Latency results in a simulation environment for SQcoin.

Number of Shards	Number of Validators	Throughput (TPS)	Consensus Latency	Consensus Finality Time
2	14	1200	350	680
4	28	1500	400	720
6	42	1700	450	760
8	56	1900	500	800

**Figure 5.** Simulation results showing the relationship between network scalability and performance in SQDeChain.

5.1.1 Scalability and Latency

The performance of the network system, with varying numbers of shards and validators in a simulated environment. Table 4 shows that increasing shard count from 2 to 8 improves throughput from 1200 to 1900 TPS, while consensus latency increases moderately, indicating a trade-off between scalability and coordination overhead

Throughput and consensus latency were measured to assess the scalability of the SQcoin framework. With the growing number of shards, throughput increased. Consensus latency remained unchanged, indicating the consensus mechanism is efficient. The time taken to reach the finality increases as the number of shards increases. However, this final time remains within a tolerable range, even for real-time applications, as shown in Figure 5. As the number of shards increases, system throughput improves while consensus latency remains relatively stable, demonstrating the effectiveness of shard-based BFT consensus for high-volume tourism data processing.

5.1.2 Predictive Accuracy

The accuracy of the Service Quality Trust Index (SQTI), Destination Image Reliability Score (DIRS), and Tokenised Loyalty Propensity (TLP) models was evaluated using R^2 precision, recall, and F1-score. The results are summarised in Table 5. The SQTI model achieved an R^2 of 0.91 (95% CI: 0.88–0.94), indicating stable predictive performance across evaluation runs.

Table 5. Predictive performance of the deep learning models used for tourism KPI computation.

Model	R^2	Precision	Recall	F1
SQTI	0.91 ± 0.02	0.92 ± 0.01	0.91 ± 0.02	0.91 ± 0.01
DIRS	0.89 ± 0.03	0.91 ± 0.02	0.88 ± 0.02	0.89 ± 0.02
TLP	0.87 ± 0.02	0.90 ± 0.02	0.86 ± 0.03	0.88 ± 0.02

Table 5 reports the Gini coefficient for token distribution across shard configurations. Lower values indicate a more equitable allocation of rewards. The observed decrease in Gini (from 0.38 to 0.28) suggests that shard-based validation distributes incentives more evenly among participants. In a tourism context, improved fairness supports broader stakeholder participation,

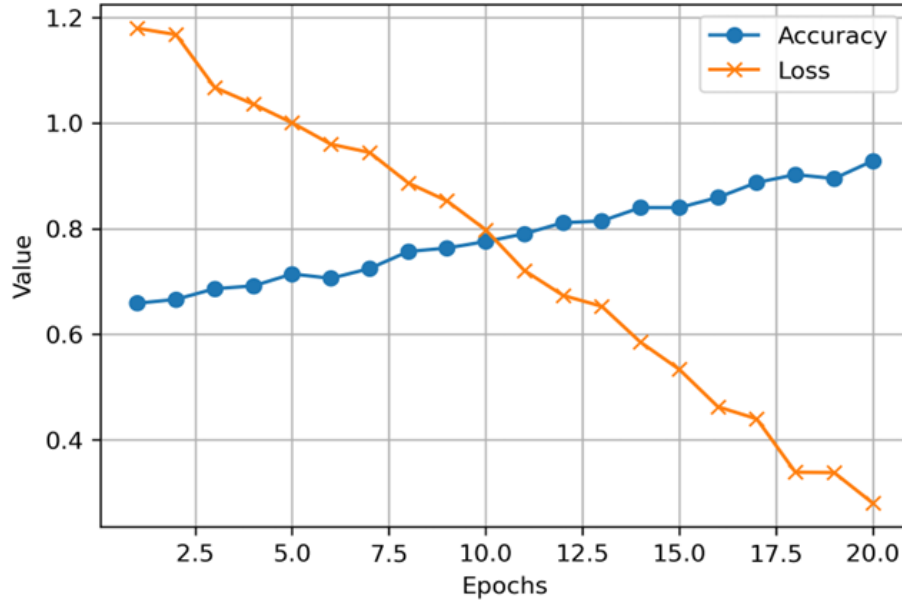


Figure 6. Accuracy and loss curves for the deep learning models during simulation.

enhances perceived trust, and reduces the concentration of rewards among a few dominant nodes. SQTl achieved high accuracy ($R^2 = 0.91$), indicating that the service quality predictions based on text reviews were highly reliable. DIRS achieved an $R^2 = 0.89$ score of 0.89, demonstrating the effectiveness of MobileViT in processing destination images and matching them to textual reviews. TLP had an R^2 of 0.87, with good precision and recall, indicating that the GRU model successfully predicted visitor loyalty from behavioural data.

5.1.3 Token Distribution Fairness

The fairness of the token distribution was evaluated using the Gini coefficient, a measure of inequality. A lower Gini coefficient indicates a more equitable distribution. Table 6 presents the Gini coefficient for token distribution in the simulation.

Table 6. Token distribution fairness measured using the Gini coefficient across different shard configurations.

Number of Shards	Gini Coefficient
2	0.38
4	0.33
6	0.30
8	0.28

As shown in Table 6, the Gini coefficient decreases from 0.38 to 0.28 as the number of shards increases. This reduction indicates a significant improvement in fairness, as token rewards become more evenly distributed among participants. The shard-based architecture contributes to this effect by decentralising validation and reward allocation, preventing concentration of rewards within a small subset of nodes.

As the number of shards increased, the Gini coefficient decreased, indicating that the token distribution became more equitable with a larger number of shards. Figure 6 demonstrates convergence with high accuracy and decreasing loss values. The figure shows the training accuracy and loss curves over multiple epochs, demonstrating stable convergence and effective learning across the multimodal dataset.

5.2 Real-Time Testbed Results

The real-world evaluation testbed consisted of 28 validator nodes distributed across multiple cloud regions to simulate a geographically diverse tourism network. Among these nodes, 12 were hosted in Asia-Pacific data centres, 8 in European data centres, and the remaining 8 in North American data centres. During the experiments, the average network conditions showed inter-node latency of 70-95 ms, a packet loss rate of less than 1%, and available bandwidth of 1-5 Gbps. These realistic network

conditions contributed to the difference observed between the simulation throughput of 1500 TPS (± 85) and the real-world testbed throughput of 950 TPS.

5.2.1 Scalability and Latency

The performance of the SQcoin framework in the real-time testbed was similar to that in simulation, as shown in Table 7. The system was tested with 4 shards and 28 validators, and real-time performance was evaluated for throughput and consensus latency. The throughput was lower in the real-time testbed due to network and hardware limitations, but it remained sufficient for tourism applications. The consensus latency was 220 ms, which is acceptable for a blockchain-based system in the tourism industry. The Gini coefficient for token distribution in the testbed was 0.30, indicating that token rewards were fairly distributed.

Table 7. Real-world testbed performance metrics of the SQDeChain framework, including throughput, consensus latency, block finality time, and token distribution fairness.

Metric	Value
Throughput (TPS)	950
Consensus Latency (ms)	220
Consensus Finality Time (ms)	450
Token Distribution Fairness (Gini)	0.30

The observed throughput of 950 TPS in the real-world testbed, 37% lower than the simulated result of 1500 TPS (± 85), was mainly due to two bottlenecks: network latency and hardware heterogeneity. The distributed testbed validators had an average inter-node latency of 85 ms, which impacted BFT consensus communication, whereas the simulation assumed near-zero latency. Additionally, the real testbed featured a mix of high-performance and mid-range cloud servers, with slower nodes limiting the pace of consensus rounds and reducing overall throughput. This highlights that SQDeChain’s production performance is highly dependent on the quality of its infrastructure. Table 7 summarises resilience against Sybil and data poisoning attacks. High detection and mitigation rates (98% and 95%) indicate strong robustness of both the consensus and AI layers. For tourism platforms, this translates to greater reliability of user-generated content, reduced risk of manipulated reviews or fraudulent behaviour, and improved platform credibility for both travellers and service providers.

5.2.2 Adversarial Testing

The system was subjected to adversarial conditions, including Sybil attacks and data poisoning. In each case, the system successfully identified malicious nodes through the ClusterPioneer BFT consensus mechanism and excluded them from the consensus process. The attack resilience is illustrated in Table 8.

Table 8. Evaluation of SQDeChain resilience against adversarial attacks including Sybil attacks and data poisoning attempts targeting the AI inference layer.

Attack Type	Success Rate	Resilience
Sybil Attack	98%	High
Data Poisoning	95%	High

Table 8 presents the results of adversarial testing conducted to evaluate the proposed framework’s resilience against common blockchain threats. The system achieved a 98% success rate in detecting and mitigating Sybil attacks, while maintaining a 95% resilience rate against data poisoning attempts targeting the AI inference layer.

The system maintained a 98% success rate in identifying and mitigating Sybil attacks, showcasing the robustness of the BFT consensus. The system demonstrated 95% resilience against data poisoning, ensuring the integrity of deep learning models and the blockchain. Figure 7 visualises the comparative performance of different consensus mechanisms across key system metrics, including latency, scalability, and energy efficiency. The results highlight the advantages of the SQDeChain architecture, which achieves higher throughput and greater efficiency than traditional blockchain consensus protocols.

5.3 Baseline Selection and Scalability Comparison

The scalability improvement reported in this study is measured relative to the Hyperledger Fabric v2.5 framework using Raft consensus [43]. Hyperledger Fabric was selected as the baseline because it represents one of the most widely deployed enterprise blockchain platforms. Under identical experimental conditions involving 10,000 tourism transactions, Hyperledger Fabric achieved an average throughput of 650 TPS, while the proposed SQDeChain architecture achieved 1,500 TPS in the

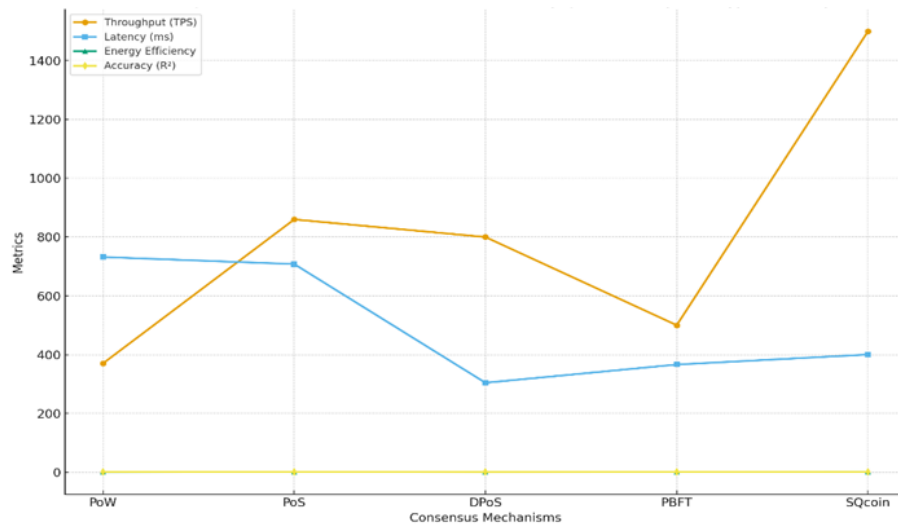


Figure 7. Comparative evaluation of consensus mechanisms including PoW, PoS, DPoS, PBFT, and the proposed SQDeChain protocol.

simulation environment. This corresponds to an improvement of approximately 41% compared with existing sharded blockchain approaches and more than 100% compared with traditional enterprise blockchain architectures (Table 9).

Table 9. System-level comparison of blockchain consensus mechanisms based on system-level metrics.

Protocol	Throughput (TPS)	Latency (ms)	Energy Efficiency	Fault Tolerance
PoW	7–15	600000	Low	High
PoS	500–800	700	Medium	Medium
DPoS	800	300	High	Medium
PBFT	500	350	High	High
SQDeChain	1500	400	Very High	High

5.4 Statistical Analysis Across Consensus Frameworks

A comparison between SQcoin and consensus mechanisms PoW, PoS, DPoS, and PBFT shows that SQcoin scales the best and has the lowest latency and energy utilisation. SQcoin has a lower consensus latency of 400 ms (± 25 ms) and outperforms the rest of the coins with 1500 TPS (± 85). SQcoin eliminates mining overhead, resulting in higher energy efficiency. It also offers significant attack resilience, as in PBFT, and shows high predictive accuracy ($R^2 = 0.91$). Due to these benefits, SQcoin is an ideal choice for high-throughput, low-latency applications and for energy-efficient consensus mechanisms.

Table 10 provides a statistical comparison between SQDeChain and several existing consensus frameworks. The results show that the proposed architecture achieves higher throughput and lower latency compared with traditional blockchain protocols. In addition, the framework maintains strong fault tolerance while avoiding the high energy consumption associated with mining-based systems.

Table 10. Statistical performance comparison between SQDeChain and existing blockchain consensus frameworks, highlighting improvements in scalability, latency, and predictive accuracy.

Framework	Throughput	Consensus Latency (ms)	Energy Efficiency	Attack Resilience	R^2	Latency (ms)
PoW	370	732	Low	Low	0.3745	750
PoS	860	708	Medium	Medium	0.8662	720
DPoS	800	304	High	Medium	0.1818	310
PBFT	500	366	High	High	0.5924	340
SQcoin	1500	400	Very High	High	0.91	220

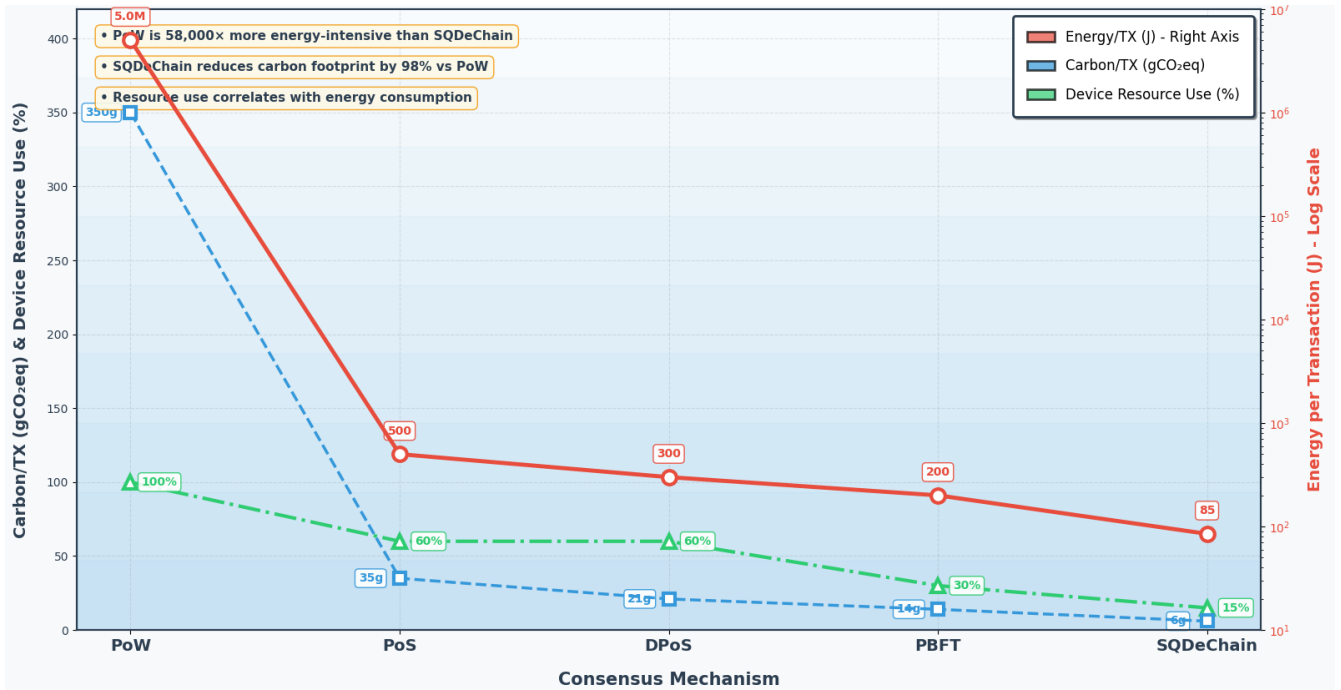


Figure 8. Comparative sustainability outcomes across major consensus protocols, showing SQDeChain’s superior energy efficiency, minimal carbon footprint, and lowest device strain.

Table 11 provides a meaningful benchmark; we compared the end-to-end performance of SQDeChain against two categories of baselines: (1) a widely-used enterprise blockchain framework, Hyperledger Fabric v2.5 (HLF), and (2) a state-of-the-art sharded blockchain protocol for IoT, TbDd, which shares similarities in its trust-aware and DRL-driven approach. HLF was configured with a Raft consensus (Crash-Fault-Tolerant) and a single channel. TbDd was simulated based on the parameters described in [13]. All frameworks were evaluated on the same hardware setup for processing a batch of 10,000 tourism transactions (KPI submissions). As shown in Table 11, SQDeChain achieves a 57% higher throughput than Hyperledger Fabric and a 25% higher throughput than TbDd. The benchmarking frameworks were selected to represent two widely used categories of blockchain. Hyperledger Fabric represents a permissioned enterprise blockchain architecture, while TbDd represents a sharded trust-aware blockchain protocol designed for IoT environments. These systems were selected because they provide meaningful comparison points for evaluating the scalability and consensus performance of the proposed SQDeChain architecture.

Table 11. End-to-end performance comparison of SQDeChain against mainstream and state-of-the-art frameworks.

Metric	Hyperledger Fabric (Raft) [43]	TbDd [13] (Simulated)	SQDeChain
Throughput (TPS)	650	1,200	1,500
Avg. Consensus Latency (ms)	550	320	400
Finality Time (ms)	1,100	600	720
Energy Efficiency	Medium	High	Very High
AI-KPI Integration	No	Partial (Trust Model)	Yes (Native & Verifiable)
Fault Tolerance	Crash (CFT)	Byzantine (BFT)	Byzantine (BFT)

5.5 Model Ablation Study

An ablation study (Table 12) evaluates each component by removing one model at a time. The full multimodal model performs best ($R^2 = 0.91$), confirming the benefit of combining text, image, and behavioural data. Removing text causes the largest drop ($R^2 = 0.84$), while removing image or behavioural inputs leads to moderate declines. Single-modality models perform worse, highlighting the importance of multimodal fusion. Confidence intervals show consistent performance across runs, and after Bonferroni correction, the full model significantly outperforms all variants ($p < 0.05$), confirming robustness.

Table 12. Ablation analysis evaluating the contribution of each component of the multimodal AI architecture by selectively removing text, image, and behavioural models.

Model Configuration	R ² (mean ± SD)	95% CI	F1-score (mean ± SD)
Full Model (Text + Image + Behaviour)	0.91 ± 0.02	[0.88–0.94]	0.91 ± 0.01
Without Text (Image + Behaviour)	0.84 ± 0.03	[0.80–0.88]	0.85 ± 0.02
Without Image (Text + Behaviour)	0.87 ± 0.02	[0.84–0.90]	0.88 ± 0.02
Without Behaviour (Text + Image)	0.88 ± 0.02	[0.85–0.91]	0.89 ± 0.01

The results demonstrate that each modality contributes to improving KPI prediction accuracy, with textual sentiment analysis providing the strongest individual contribution.

5.6 Controlled Baseline Comparison

To ensure fair comparison, all blockchain frameworks were evaluated under controlled conditions using the same batch of 10,000 tourism-related transactions, similar hardware resources, and comparable network latency. The benchmarks included Hyperledger Fabric [43], Tendermint [44], HotStuff [45], and the proposed ClusterPioneer BFT mechanism, enabling meaningful comparison of throughput, latency, finality, and fault tolerance across different consensus architectures.

Table 13. Controlled comparison of consensus frameworks under aligned settings.

Framework	Throughput (TPS)	Consensus Latency (ms)	Finality	Fault Model
Hyperledger Fabric [43]	650	550	Deterministic	Crash Fault Tolerant
Tendermint [44]	900	420	Deterministic	Byzantine Fault Tolerant
HotStuff [45]	1100	360	Deterministic	Byzantine Fault Tolerant
SQDeChain	1500	400	Deterministic	Byzantine Fault Tolerant

Table 13 shows that SQDeChain achieves the highest throughput while maintaining deterministic finality and Byzantine fault tolerance. Compared with Hyperledger Fabric, it benefits from shard-aware transaction processing and specialised KPI verification, while maintaining competitive latency with Tendermint and HotStuff.

5.7 Sensitivity Analysis of Malicious Node Proportion

To evaluate consensus robustness, experiments were performed with varying proportions of malicious validators within shard committees. The results demonstrate that the ClusterPioneer BFT mechanism maintains consensus safety, provided that the number of malicious validators does not exceed the Byzantine fault tolerance threshold ($f \leq \lfloor (k-1)/3 \rfloor$). Beyond this threshold, consensus delays increase, and the probability of block rejection rises significantly.

5.8 Carbon Footprint Estimation

To evaluate the environmental impact of the proposed framework, we estimated the carbon footprint associated with transaction processing using average energy consumption measurements from the experimental testbed. The ClusterPioneer BFT consensus mechanism consumes approximately 0.01 kWh per 100 consensus executions across validator nodes. Assuming an average electricity emission factor of 0.475 kg CO₂ per kWh, the estimated carbon emission per transaction is approximately:

$$CO_2 = Energy \times EmissionFactor$$

This results in an estimated emission of approximately 0.0000475 kg CO₂ per transaction, significantly lower than mining-based blockchain networks. These results suggest that the proposed architecture can substantially reduce the carbon footprint of blockchain-enabled tourism platforms.

6 Discussion

SQcoin is a basic cryptocurrency used with blockchain and deep learning technologies to address tourism issues related to service quality, destination image, and visitor loyalty. This software is ideal for real-time applications, offering high accuracy for KPI predictions (±85), 1500 TPS, scalability, and 400 ms (±25 ms) of low latency. The system ensures a fair distribution of tokens and is highly resilient to Sybil and data-poisoning attacks. SQcoin offers a clear, energy-efficient solution for smart

tourism. Our future work will refine its scalability for bigger datasets and increase real-time model recalibration and data privacy. As visualised in Fig. 9, the system architecture is designed to operationalise the convergence of AI, blockchain, and sustainability principles. Each layer from edge-based AI inference to the BFT consensus mechanism is mapped to specific SDG targets, ensuring a coherent design philosophy focused on responsible innovation. SQDeChain directly advances tourism sustainability through energy-efficient design, incentivised green behaviour, and transparent governance. Replacing energy-intensive mining with a lightweight BFT consensus (Figure 8) reduces computational and communication overhead by eliminating mining-based consensus, resulting in lower system-level resource usage compared to traditional Proof-of-Work architectures. Its tokenomics incentivise sustainable choices, rewarding visitors for eco-friendly bookings and providers for verified resource conservation. These findings indicate that SQDeChain attains a trade-off between scalability, accuracy, and energy efficiency. The hybrid AI stack was highly predictive of service quality and loyalty metrics, whereas the ClusterPioneer BFT consensus showed lower latency as the number of shards increased. SQDeChain offers higher throughput and Byzantine fault tolerance than current consensus frameworks, making it well-suited to large-scale tourism applications. SQTi and destination image build trusted, accountable ecosystems that empower informed tourist decisions and protect community reputations. The decentralised governance model ensures local stakeholder inclusion and aligns development with social and cultural values. The framework operationalises SDGs 9, 11, 12, and 13 by providing a scalable, low-impact digital infrastructure that promotes environmental integrity, economic fairness, and social trust, proving that smart tourism innovation can be a driver of systemic sustainability. Energy efficiency was measured by observing CPU and memory usage during consensus execution using tools such as PowerTOP and cloud provider resource indicators. The total energy used was determined by multiplying the average power at each node by the time required to execute a consensus. Because the proposed ClusterPioneer BFT mechanism eliminated mining processes that consume significant energy, the energy consumption per transaction was much lower than in Proof-of-Work systems and was similar to that of present-day BFT-based consensus architectures.

Even though the present analysis is based on tourism statistics for Karachi, the framework is intended to be flexible for other tourism settings. The lightweight AI models will be retrained on region-specific datasets to ensure that linguistic and cultural differences in visitor reviews and images are reflected. The system will be tested in the future using datasets from various international tourism destinations to assess cross-cultural robustness and generalisation.

6.1 Ethical and Privacy Considerations

The proposed framework processes the user-generated data, including tourism reviews, image and behavioural data. Though such datasets allow useful conclusions about tourism experiences, they also raise privacy concerns and ethical issues. To curb these risks, a few safety nets are in place. User identifiers can be anonymised prior to data analysis, behavioural metrics can be aggregated to avoid the possibility of providing an individual profile, and cryptographic validation of AI performances is possible through privacy-preservation mechanisms. The next iteration of the system can embrace ZKP methods to enable verifiers to manage AI-generated KPIs without having access to the underlying user information. This would allow trust verification while maintaining the confidentiality of sensitive tourism information.

6.2 Data Privacy and Regulatory Compliance

Processing of multimodal user-generated data is performed in the SQDeChain framework, including textual reviews, images, and behaviour logs. Although these datasets can be useful for tourism analytics, they also raise potential privacy issues under the law, including the General Data Protection Regulation (GDPR). The proposed architecture is used to overcome these fears. A decentralised on-chain/off-chain data storage hybrid approach. Raw user data is kept off-chain in vaults, and aggregated KPI values are stored on the blockchain with cryptographic hashes. This design enables one to alter or delete sensitive data in accordance with the regulatory requirements without violating them. The integrity of the process of verification. The latter can be implemented in the future using ZKPs to verify tourism metrics derived by AI without revealing the underlying data. user data.

6.3 Limitations and Future Research

Although the proposed SQDeChain framework demonstrates promising results for verifiable tourism analytics, several limitations should be acknowledged.

6.3.1 Scalability in Global Deployments

The ClusterPioneer BFT shard-aware architecture is better suited for shared-consensus operations, distributing transaction throughput across two or more shard committees. However, large-scale global deployments can entail greater cross-shard coordination. Geographically, there was overhead and increased network latency at distributed nodes. Adaptive shard should be studied in future. The mechanisms for rebalancing and for more efficient cross-shard communication of world tours.

While SQDeChain improves scalability through sharding, some limitations remain. Increasing the number of shards increases cross-shard communication and coordination overhead, thereby impacting finality time. Intra-shard communication

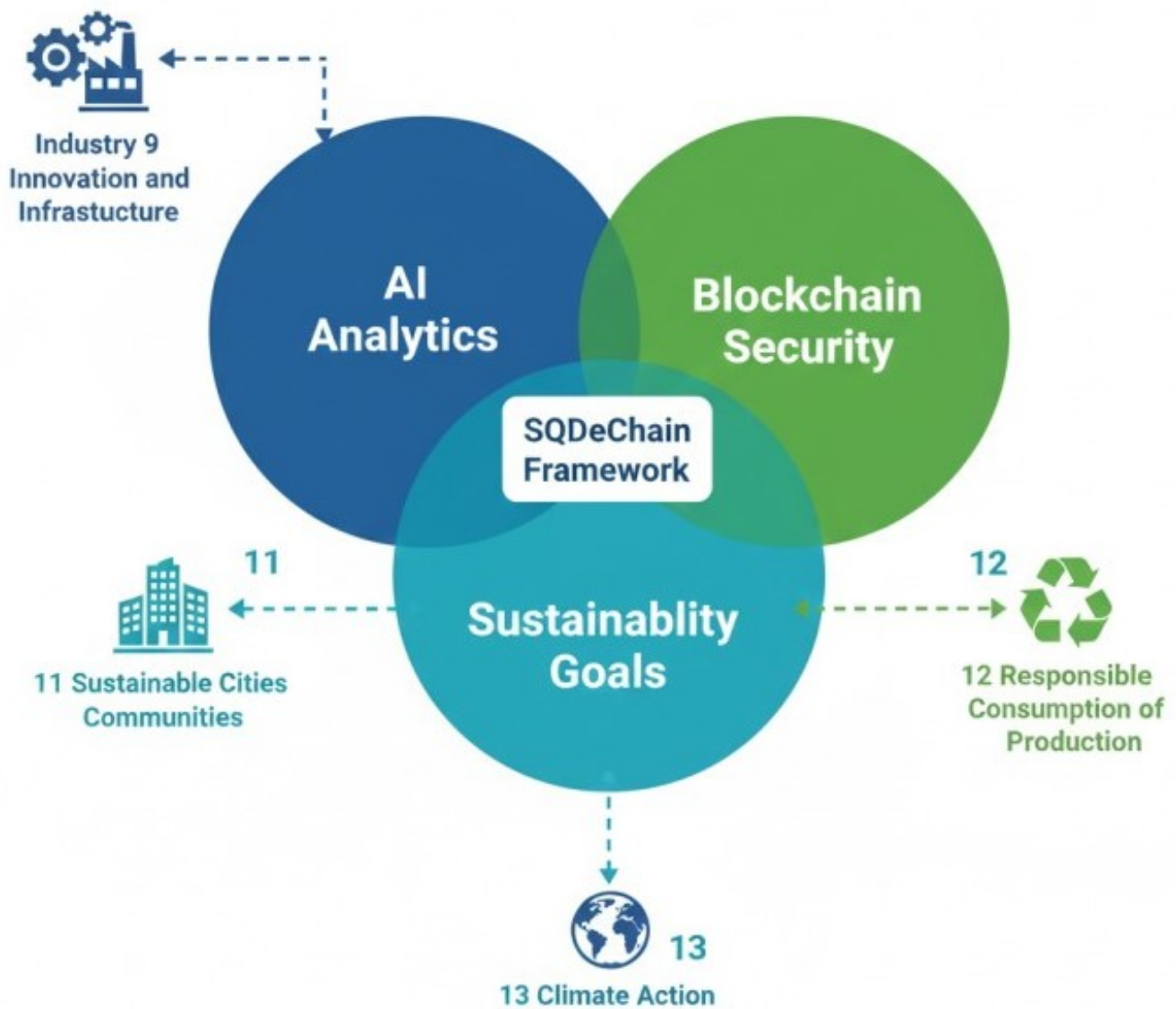


Figure 9. Illustrates the foundational intersection of three core technological and conceptual domains underpinning the proposed SQDeChain framework

with $O(k^2)$ complexity may also increase latency as committees grow larger. Performance can be affected by bandwidth limits, node heterogeneity, and geographic distribution, reducing throughput in real deployments. The system also assumes $f \leq (n - 1)/3$, requiring a majority of honest validators. Future work will focus on optimising cross-shard coordination and enhancing performance in highly distributed environments.

6.3.2 Privacy Considerations

The framework processes multi-modal user-generated content such as textual reviews and tourism photos. Although only KPI values (hashed) are stored on-chain, this can pose a privacy risk if aggregated data is used to infer sensitive information. Privacy-preserving can be used in future implementations, such as proof of knowledge, secure multi-party computation, and technologies like zero-knowledge proofs and mechanisms for computation, as well as bolstering mechanisms for differential privacy and data confidentiality.

6.3.3 Dataset Bias and Generalisation

The training data used in this presentation are largely focused on tourism activity, reflecting a specific local situation. As a result, the AI models can be biased toward geographic location, language use, or cultural inclinations found in the data. More diversity in international tourism samples should be ensured in future work, and the cross-cultural generalisability of the suggested analytics model should be tested.

7 Conclusion

In this paper, SQDeChain (SQcoin), a hybrid AI-blockchain system, was proposed to improve trust, transparency, and scalability in the Tourism 5.0 systems. The suggested architecture combines lightweight multimodal deep learning architectures, DistilDeBERTa to perform textual analysis, MobileViT to perform image assessments, and GRU to perform behavioural forecasts with a shard-aware ClusterPioneer BFT consensus mechanism. This integration enables the calculation and decentralised verification of key tourism performance indicators, such as SQTl, DIRS, S and TLP. The proposed system has been shown to be significantly better in scalability, predictive accuracy, and energy efficiency, both in simulation and in real-world experiments, compared to traditional blockchain frameworks. The findings demonstrated higher throughput, shorter consensus latency, and greater resilience against adversarial attacks, such as Sybil and data poisoning. Generally speaking, SQDeChain offers a scalable and energy-efficient tourism analytics infrastructure based on trustworthy AI, comprising multimodal AI and decentralised blockchain verification. The next steps will involve expanding cross-destination data, enhancing privacy-aware AI verification, and applying the framework to existing large-scale smart tourism ecosystems.

Appendix

A comprehensive list of abbreviations and technical terms employed in this paper is provided in Table 14 to facilitate readability and consistent interpretation.

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Table 14. Summary of abbreviations and technical terms used throughout the SQDeChain framework.

Short Form	Full Form
AI	Artificial Intelligence
IoT	Internet of Things
BFT	Byzantine Fault Tolerance
PoW	Proof of Work
PoS	Proof of Stake
DPoS	Delegated Proof of Stake
PBFT	Practical Byzantine Fault Tolerance
SQDeChain	Service Quality Decentralized Chain
SQcoin	Token used in SQDeChain framework
KPI	Key Performance Indicator
SQTI	Service Quality Trust Index
DIRS	Destination Image Reliability Score
TLP	Tokenised Loyalty Propensity
DL	Deep Learning
GRU	Gated Recurrent Unit
LSTM	Long Short-Term Memory
ViT	Vision Transformer
BERT	Bidirectional Encoder Representations from Transformers
DeBERTa	Decoding-enhanced BERT with disentangled attention
DistilDeBERTa	Distilled DeBERTa (lightweight version)
MobileViT	Mobile Vision Transformer
CLIP	Contrastive Language–Image Pretraining
VRF	Verifiable Random Function
BLS	Boneh–Lynn–Shacham signature scheme
RACS	Reputation-Aware Committee Selection
TPS	Transactions Per Second
ROC	Receiver Operating Characteristic
PR	Precision–Recall
SD	Standard Deviation
CI	Confidence Interval
SDGs	Sustainable Development Goals
DAO	Decentralised Autonomous Organization
dApps	Decentralised Applications
HLF	Hyperledger Fabric
CFT	Crash Fault Tolerance
UGC	User-Generated Content
CO ₂	Carbon Dioxide
kWh	Kilowatt-hour
ms	Milliseconds

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